

V-I Characteristics Measurement of a Diode

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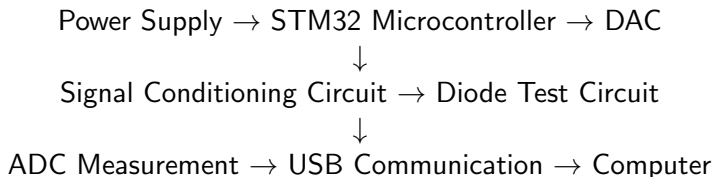
Introduction

- A diode is a semiconductor device that allows current flow primarily in one direction.
- The V-I characteristics curve represents the relationship between the voltage across the diode and the current flowing through it.
- Studying this curve helps in understanding the behavior of the diode under different biasing conditions.
- Important parameters obtained from the curve include:
 - Cut-in (threshold) voltage
 - Forward dynamic resistance
 - Reverse leakage current

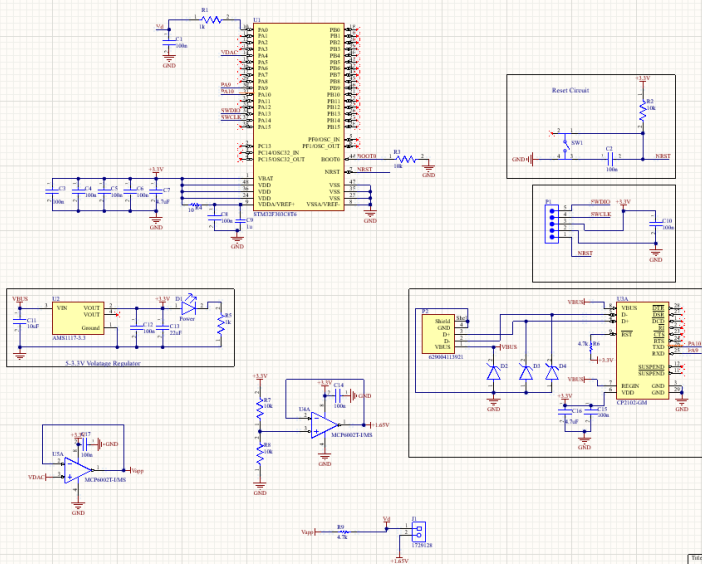
Objective

- Design and implement a compact PCB system for measuring the V-I characteristics of a diode.
- Generate a programmable voltage sweep using the DAC of a microcontroller.
- Measure the resulting voltages using the ADC.
- Compute current through the diode using a sensing resistor.
- Transfer measurement data to a computer for analysis and visualization.

System Block Diagram



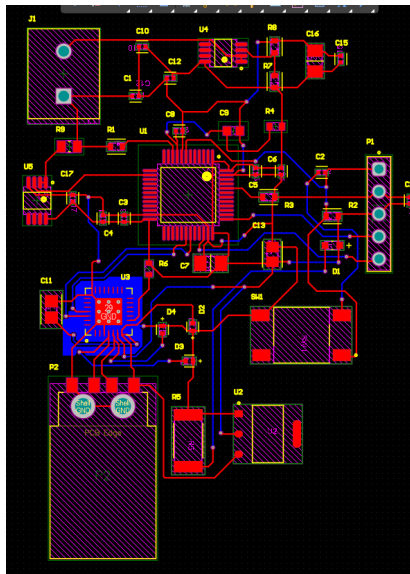
Circuit Schematic



Circuit Schematic

- The schematic integrates the microcontroller, power supply, analog circuitry and communication interface.
- The DAC generates the voltage sweep applied to the diode circuit.
- The ADC measures voltages required for current calculation.

PCB Layout



- The PCB was designed using Altium Designer.
- A two-layer PCB structure was used.
- Ground plane was implemented to reduce electrical noise.
- Analog and digital sections were routed carefully to ensure measurement accuracy.

Power Supply Section

- The system is powered using a 5V supply from the USB interface.
- A linear regulator (AMS1117-3.3) converts the 5V input to a stable 3.3V supply.
- The regulated 3.3V powers:
 - STM32 microcontroller
 - Operational amplifier
 - USB communication IC
 - Analog circuitry
- Decoupling capacitors are placed close to IC power pins to reduce noise.

- STM32F303CCT6 microcontroller is used as the main controller.
- It contains built-in peripherals such as:
 - 12-bit DAC for voltage generation
 - 12-bit ADC for voltage measurement
 - UART for communication
- The microcontroller generates the voltage sweep and processes measurement data.

ST-Link Programming Interface

- Firmware is uploaded using an ST-Link debugger.
- Programming is performed through the Serial Wire Debug (SWD) interface.
- Required signals:
 - SWDIO – Data line
 - SWCLK – Clock signal
 - NRST – Reset
 - GND – Ground
 - 3.3V – Target voltage reference

USB Communication Interface

- USB communication between the computer and the microcontroller is implemented using the CP2102-GM USB-to-UART bridge IC.
- The CP2102-GM converts USB signals from the computer into UART serial communication for the microcontroller.
- This allows measurement data such as diode voltage and current to be transmitted from the system to the computer.
- The computer detects the device as a virtual COM port, enabling easy data acquisition using serial communication software.
- This interface simplifies communication and allows real-time monitoring of the diode V-I characteristics.

Firmware Workflow

- 1 Initialize DAC, ADC and UART peripherals.
- 2 Generate a voltage sweep using the DAC.
- 3 Apply the voltage to the diode measurement circuit.
- 4 Measure voltage using the ADC.
- 5 Calculate current using Ohm's law.
- 6 Transmit measurement data to the PC through USB.
- 7 Plot the V-I curve using a Python script.

Signal Conditioning Circuit

- An operational amplifier (MCP6002) is used for buffering.
- The DAC generates voltages between 0 and 3.3V.
- A reference voltage of 1.65V is generated using a resistor divider.
- This reference allows the circuit to observe both forward and reverse diode behavior.

Diode Measurement Circuit

- The diode under test is connected in series with a sensing resistor.
- The DAC voltage sweep is applied across this circuit.
- Voltage across the sensing resistor is measured using the ADC.

Current through the diode is calculated using:

$$I = \frac{V_{resistor}}{R_{sense}}$$

Working Principle

- 1 Microcontroller generates a controlled voltage sweep.
- 2 The voltage is applied to the diode circuit.
- 3 ADC measures the voltages in the circuit.
- 4 Current is calculated using Ohm's law.
- 5 Data is transmitted to the PC.
- 6 The V-I curve is plotted using software.

Bill of Materials (BOM)

- Microcontroller: STM32F303CCT6
- Voltage Regulator: AMS1117-3.3
- USB-UART Converter: CP2102N
- Operational Amplifier: MCP6002
- Sensing Resistor
- Biasing Resistors
- Decoupling Capacitors
- SWD Programming Header
- USB Type-B Connector
- Diode Test Connector

Expected V-I Characteristics

- Forward current increases exponentially after the cut-in voltage.
- Silicon diodes typically conduct near 0.6V to 0.7V.
- Reverse bias shows very small leakage current.
- The curve demonstrates nonlinear semiconductor behavior.

Conclusion

- A PCB-based automated system for diode V-I measurement was designed.
- STM32 DAC and ADC were used for voltage generation and measurement.
- USB communication enables easy transfer of data to the computer.
- The system provides a compact platform for semiconductor device characterization.